



Prop

Let X be any space and Y be Compact Hausdorff space.

$f: X \rightarrow Y$ is continuous map if and only if the graph $G_f = \{(x, f(x)) \mid x \in X\}$ is closed.

Proof. Let $f: X \rightarrow Y$ be a continuous map.

Let $(x, y) \in X \times Y \setminus G_f$ be given. Then, $y \neq f(x)$, so there exist disjoint open sets U, V

$$U \cap V = \emptyset, \quad y \in U, \quad f(x) \in V$$

Since $f(x) \in V \stackrel{\text{def}}{\Leftrightarrow} x \in f^{-1}[V]$ and f is continuous, $f^{-1}[V]$ is an open set in X .

$$\text{Now, } (x, y) \in f^{-1}[V] \times U$$

And, $(f^{-1}[V] \times U) \cap G_f = \emptyset$, because if $(x, f(x)) \in f^{-1}[V] \times U$, then

$f(x) \in V$ and $f(x) \in U$, it is contradiction to disjointness.

Therefore, $(x, y) \in f^{-1}[V] \times U \subseteq X \times Y \setminus G_f$, so proved. $\Rightarrow$

Conversely, let G_f is closed in $X \times Y$, under the Product Topology.

Let $C \subseteq Y$ be a closed set in Y . Then, $G_f \cap (X \times C)$ is closed. Moreover,

$$P_X[G_f \cap (X \times C)] = \{x \in X \mid f(x) \in C\} = f^{-1}[C]$$

is closed, because $P_X: X \times Y \rightarrow X: (x, y) \mapsto x$ is closed map if Y is compact. $\square$

Note: for $g = f \times i: X \times Y \rightarrow Y \times Y: (x, y) \mapsto (f(x), y)$,

$$g^{-1}[\Delta(Y)] = \{(x, y) \in X \times Y \mid (f(x), y) \in \Delta(Y)\} = G_f$$

Prop.

Compact Hausdorff space is Normal space.